

Introduction to **DAX (Data Analysis Expressions)**



1. What is DAX?

DAX (Data Analysis Expressions) is a formula language used in Power BI, Power Pivot, and SSAS Tabular.

It is used to:

- Create Calculated Columns
- Create Measures
- Create Calculated Tables
- Perform aggregations
- Time Intelligence
- Ranking
- Filtering
- Business calculations



2. Why DAX?

Suppose we have this Sales table.

Product	Qty	Price	Total Amount
Pen	5	10	50
Book	2	100	200
Bag	1	800	800

$\text{Total Amount} = \text{Sales[Qty]} * \text{Sales[Price]}$



Types of DAX Objects

1. Calculated Column:

Creates a value for every row.

2. Measure:

Calculates dynamically depending on filters.

Example:

Total Sales = SUM(Sales[Amount])

3. Calculated Table:

Creates an entirely new table.

Example:

TopCustomers =

FILTER(Customers, Customers[Sales] > 50000)



3. Common DAX Functions

Mathematical:

- SUM()
- AVERAGE()
- MIN()
- MAX()
- COUNT()
- DISTINCTCOUNT()

Logical:

- IF()
- SWITCH()
- AND()
- OR()

Example:

Result =
IF(Sales[Amount] > 1000,
"High",
"Low")

Text:

- LEFT()
- RIGHT()
- MID()
- LEN()
- CONCATENATE()
- UPPER()
- LOWER()

Date:

- YEAR()
- MONTH()
- DAY()
- TODAY()
- DATEDIFF()

Aggregate:

- SUMX()
- AVERAGEX()
- COUNTX()

Filter Functions:

- FILTER()
- ALL()
- VALUES()
- RELATED()
- LOOKUPVALUE()



Introduction to M Language



1. What is M Language?

M Language is the scripting language used inside Power Query Editor.

Whenever we clean or transform data, Power Query automatically writes M code.

We don't need to write M manually for most tasks.



2. Where is M Language Used?

Only inside Power Query Editor before the data is loaded into the data model.

Typical uses include:

- Data cleaning
- Data transformation
- Data preparation
- ETL (Extract, Transform, Load)
- Data reshaping

Key takeaway:

Use M Language to prepare and shape the data before it is loaded. Use DAX to analyze that prepared data and create interactive business insights after it is loaded. This separation leads to cleaner models, better performance, and easier maintenance.



DAX vs M Language

Feature	DAX	M Language
Works In	Data Model	Power Query
Purpose	Analysis & Calculations	Data Cleaning & Transformation
Executed	After data is loaded	Before data is loaded
Used For	Measures, Calculated Columns, Calculated Tables	Cleaning, Merging, Splitting, Filtering, Shaping
Performance	Runs during report interaction	Runs during data refresh
Target	Business calculations	ETL and data preparation

